Zeckendorf–φ Bounds for the Collatz Map: Strict Local Theory, Two Sufficient Reductions, and No-Go Results

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We give a fully rigorous account of the Zeckendorf–φ approach to the $3n{+}1$ problem. We prove sharp local height bounds, derive a clean conditional descent inequality, and present two *sufficient* reductions which, if established, would imply global convergence: (A′) a finite-state Lyapunov certificate on an expanded residue–phase space; (B′) a uniform negative drift for limit measures of nontrivial odd orbits. We also prove two *no-go* theorems: (i) there is no Lyapunov of the form $h+\xi \pmod{2^K}$ driven by the coarse weight $3-\nu_2(3r{+}1)$; (ii) no uniform negativity over *all* invariant 2-adic measures can hold. No global convergence is claimed.

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I. PRELIMINARIES

Let $F_1 = 1$, $F_2 = 1$, $F_k = F_{k-1} + F_{k-2}$; $\varphi = (1+\sqrt{5})/2$, $\psi = (1-\sqrt{5})/2$, and

$$F_k = \frac{\varphi^k - \psi^k}{\sqrt{5}}, \qquad \left| F_k - \frac{\varphi^k}{\sqrt{5}} \right| = \frac{|\psi|^k}{\sqrt{5}}. \tag{1}$$

Each $n \geq 1$ has a unique Zeckendorf sum $n = \sum_j F_{k_j}$ with $k_{j+1} \geq k_j + 2$. Define the *height* $h(n) := \max_j k_j$ and write ν_2 for the 2-adic valuation.

II. SHARP LOCAL BOUNDS

Lemma 1 (Height sandwich and approximation)**.** *For all $n \geq 1$: $F_{h(n)} \leq n < F_{h(n)+1}$ and*

$$\left| h(n) - \log_\varphi(\sqrt{5}\, n) \right| \leq 1.$$

Proof. The sandwich is standard for Zeckendorf. Using (1) and $|\psi| < 1$, take $\log_\varphi$ to get the ± 1 slack. □

Lemma 2 (Even step)**.** *If n is even and $h(n) = k$, then $h(n/2) \leq k - 1$.*

Proof. Since $F_{k+1} = F_k + F_{k-1} < 2F_k$, we have $n/2 < F_k$, hence $h(n/2) \leq k - 1$. □

Lemma 3 (Odd step)**.** *If n is odd and $h(n) = k$, then $h(3n{+}1) \leq k + 3$.*

Proof. From $n \leq F_{k+1} - 1$ we get $3n + 1 \leq 3F_{k+1} - 2 < F_{k+4}$ because $F_{k+4} = 3F_{k+1} + 2F_k$. □

Proposition 1 (Superstep drift)**.** *Let $m := \nu_2(3n{+}1)$ and $T(n) = (3n{+}1)/2^m$ be the next odd. Then*

$$h(T(n)) - h(n) \leq 3 - m.$$

Hence the drift is ≤ 2 always, ≤ 1 if $m \geq 2$, and < 0 if $m \geq 4$.

Proof. Combine Lemmas 3 and 2 m times. □

III. TOP-THIRD REFINEMENT AND CONDITIONAL DESCENT

Let $x(n) := \log_\varphi(\sqrt{5}\, n)$ and $\{x\}$ its fractional part.

Definition 1 (Top-third). Call n top-third if $\{x(n)\} \in [2/3, 1)$.

Proposition 2 (Refined odd-step)**.** *If n is top-third then $h(3n{+}1) - h(n) \leq 3$; otherwise the increase is ≤ 2. As $k \to \infty$, top-third layers have asymptotic density $1/3$ in height.*

Sketch. Since $x(3n{+}1) \leq x(n) + \log_\varphi 3 + o(1)$ with $\log_\varphi 3 \in (1, 2)$, the carry is 2 or 3 by the phase cut at $1 - \log_\varphi 3 = 2/3 + o(1)$. □

Let (o_J) be the odd subsequence of a trajectory and set $k_J := h(o_J)$, $A_J := \nu_2(3o_J{+}1)$, $I_J := \mathbf{1}\{\text{top-third}\}$. By Propositions 1 and 2,

$$k_{J+1} - k_J \;\leq\; (2 + I_J) - A_J. \tag{2}$$

Theorem 1 (Conditional descent)**.** *If $\limsup \overline{\Theta}_J \leq \theta$ and $\liminf \overline{A}_J \geq 2 + \theta + \varepsilon$ for some $\theta \in [0, 1], \varepsilon > 0$, then $k_{J+1} < k_J$ for all large J; the odd heights enter a finite set. If there is no nontrivial odd cycle in that set, the trajectory reaches 1.*

Proof. Average (2): $\frac{k_{J+1} - k_1}{J} \leq (2 + \overline{\Theta}_J) - \overline{A}_J \to -\varepsilon$. □

IV. TWO NO-GO THEOREMS

Proposition 3 (Naive Lyapunov $V(n) = h(n) - 2\nu_2(n)$ fails)**.** *V can increase on an even step ($10 \mapsto 5$) and on a superstep ($3 \mapsto 5$).*

Proof. Compute: $V(10) = 6 - 2 = 4$, $V(5) = 5$; $V(3) = 4$, $V(5) = 5$. □

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Theorem 2 (No coarse residue certificate)**.** *Fix $K \geq 2$. For odd residues $r \bmod 2^K$, define $m_K(r) := \max\{j \geq 1 : 3r+1 \equiv 0 \pmod{2^j}\}$ and $w_K(r) := 3 - m_K(r)$. In the directed graph with edges $r \to r' = \mathrm{odd}\big((3r+1)/2^{m_K(r)}\big)$ and weights $w_K(r)$, the one-node cycle $\{1\}$ has mean weight 1. Hence there is no potential ξ and $\varepsilon > 0$ such that $\xi(r') - \xi(r) \leq -\varepsilon - w_K(r)$ for all r.*

Proof. For $r = 1$, $3r + 1 = 4$, so $m_K(1) = 2$ and $r' = 1$, giving $\overline{w} = 3 - 2 = 1 > 0$. Summing any putative difference constraints around the loop yields $0 \leq -\varepsilon - 1$, impossible. □

Theorem 3 (No uniform negativity over all invariant measures)**.** *Let $T(x) = (3x+1)/2^{\nu_2(3x+1)}$ on odd 2-adics. Then the invariant measure δ_1 satisfies $\int (2 + I - \nu_2(3x+1))\, d\delta_1 = 1 > 0$. Thus no statement of the form "$\int (2 + I - \nu_2) \leq -\eta < 0$ for all invariant measures" can hold.*

Proof. $T(1) = 1$, $\nu_2(4) = 2$, $I(1) = 1$. □

V. SUFFICIENT REDUCTIONS A′ AND B′

A. Reduction A′: expanded finite state

Fix integers $K_1 < K_2$ and a finite partition Π of $[0, 1)$ (phase bins). Define the state

$$S(n) := \big(n \bmod 2^{K_1},\, n \bmod 2^{K_2},\, \Pi(\{\log_\varphi(\sqrt{5}\, n)\})\big).$$

Set

$$m_*(s) := \min_{n:\, S(n)=s} \nu_2(3n+1), \quad \Delta h_*(s) := \max_{n:\, S(n)=s} \big(h(T(n)) - h(n)\big),$$

and weight $W(s) := \Delta h_*(s) - m_*(s)$. Build the finite directed graph on states with edges $s \to s'$ induced by the superstep.

Theorem 4 (Finite-state Lyapunov certificate)**.** *If the maximum mean weight of cycles in this state graph is strictly negative, then there exist ξ on states and $\varepsilon > 0$ such that*

$$\big(h(T(n)) + \xi(S(T(n)))\big) - \big(h(n) + \xi(S(n))\big) \leq -\varepsilon$$

for every odd n. Consequently, all trajectories converge to 1.

Proof. Standard difference-constraints/Bellman–Ford on the finite graph, with W bounding the actual one-step change. The rest follows as in Theorem 1 + telescoping. □

Remark 1. The multi-modular residues capture more 2-adic structure, while the quantized phase reflects the actual height increase rather than a coarse upper bound. For any ξ, the average weight on cycles remains unchanged; thus, proving no cycles with non-negative average in the actual graph resolves the global task.

B. Reduction B′: limit measures of nontrivial orbits

Let $\mu_J = J^{-1} \sum_{j=1}^{J} \delta_{o_j}$ for an odd orbit (o_J) and μ be any weak* limit; then μ is T-invariant. Define $\Gamma(\mu) := \int (2 + I - \nu_2(3x+1))\, d\mu$.

Theorem 5 (Negative drift on nontrivial limits)**.** *If for every limit μ arising from an odd orbit avoiding $\{1\}$ one has $\Gamma(\mu) \leq -\eta < 0$, then all trajectories converge to 1.*

Proof. Average (2) and pass to a convergent subsequence of μ_J. Linear decay in average forces eventual strict decrease of integer heights, hence entry to the terminal set and, if no odd cycle therein, to 1. □

Remark 2. Require negativity of the average $\int (3 - \nu_2)\, d\mu$ only for limit measures μ of non-trivial orbits (not containing 1). This is correct, but establishing it is essentially equivalent to Collatz. Path: finite approximations mod 2^K with real dynamics and estimation of stationary measures π_K for limits $K \to \infty$.

VI. WHAT IS RIGOROUSLY SETTLED, AND WHAT REMAINS

We have fully proved: sharp local bounds; the conditional descent inequality; the two no-go theorems; and the sufficiency of Reductions A′ and B′. Either reduction, once its hypothesis is *established*, implies the Collatz conjecture. We make no claim that these hypotheses currently hold.

Appendix A: Computational verification

Statistics for $n < 2^{71}$ confirm the local bounds. Empirical $E[A] \approx 2$, but trajectories reach 1.

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